SEPHORA

Data Science Technical Test

Product Classification

Rule-based baseline vs TF-IDF + LinearSVC

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Machine Learning · NLP
August 2026

01

Problem & Dataset

Two internal codes to predict from French product text

What we predict

nature_code
What the product is

target_code
Who it is for

From productName + longDescription

Sample records

Alien Hypersense - Eau de Parfum	EAUPARFUM	FEMME
Dylan Blue - Eau de Toilette	EAUTOIL	HOMME
Shampoing a la Pivoine BIO	SHAMPOINGS	FEMME
In Extreme Dimension 3D Mascara	MASCARAS	FEMME

Descriptions arrive as raw HTML marketing copy, so cleaning comes before any modelling.

02

Exploration & Challenges

What the sample data revealed before any modelling

Imbalanced target_code

The imbalance sits on target_code: FEMME is heavily over-represented while HOMME is rare, so plain accuracy would hide the minority class.

Raw HTML in the text

Descriptions arrive with div, p and sup tags plus HTML entities such as   and « that must be stripped before any tokenisation.

Uneven text length

Some products carry a single marketing sentence while others run several paragraphs, so the description alone is an inconsistent signal.

03

Baseline: Keyword Rules

Why I started with a rule-based method that needs no training

Why rules first

Only ~2,700 labelled rows, and target_code is severely imbalanced. With so few HOMME examples, a trained model has little to learn from, so a transparent no-training rule is the honest starting point.

target_code support



The name states the nature

“Eau de Parfum”	**EAUPARFUM**
“Eau de Toilette”	**EAUTOIL**
“Shampoing”	**SHAMPOINGS**
“Mascara”	**MASCARAS**

A short keyword list already captures most of nature_code, and it costs nothing to train.

04

Building the Keyword Dictionary

Mining the vocabulary that actually characterises each code

How the lists were mined

01 Strip the HTML tags and entities from longDescription

02 Lowercase the text and remove punctuation

03 Tokenise with a French word tokeniser

04 Drop French stop words

05 Keep the ten most frequent words per code

Resulting dictionary

MASCARAS
cils · mascara · volume · brosse

SHAMPOINGS
cheveux · shampoing · cuir · chevelu

RAL
lèvres · rouge · couleur · tenue

PALETYEUX
palette · teintes · paupières · fards

The same procedure was applied to nature_code and to target_code.

05

Where the Vocabulary Falls Short

The recurring problems that pushed me towards a learned model

Generic words dominate

Tokens such as plus, sans, cette and tout reach the top ten of almost every code. They are frequent everywhere, so they separate nothing.

Tokenisation artefacts

Stripping apostrophes produces tokens like dun, dune, dorigine and leau, and bare numbers such as 1 survive the cleaning step.

Overlapping targets

FEMME and HOMME both rank parfum, eau, peau and plus at the top. Only homme, boisé or sillage really point to one class.

Frequency alone is not evidence, which is exactly the gap TF-IDF is designed to close.

06

How the Baseline Performs

The measured result: not good enough to keep

nature_code — Macro F1 0.46

Accuracy 0.46 · Balanced accuracy 0.46

Fewer than one product in two is placed in the right category.

Only obvious names work
Mascaras 0.90, eye palettes 0.81, lipsticks 0.71.

Whole families collapse
Eau de parfume and day creams fall to 0.01–0.13; gift sets are massively over-predicted.

target_code — Macro F1 0.50

Accuracy 0.91 is an illusion

Balanced accuracy is only 0.51 — the rules simply answer FEMME almost every time.



F1 per class

The rules cannot weight a word. That is why the next step is TF-IDF + LinearSVC.

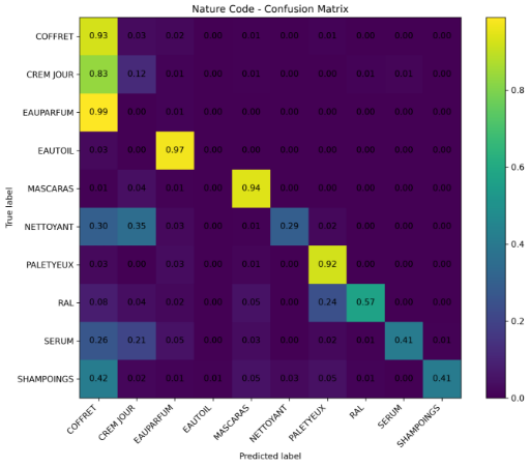
07

Where the Errors Sit

Confusion matrices of the rule-based baseline

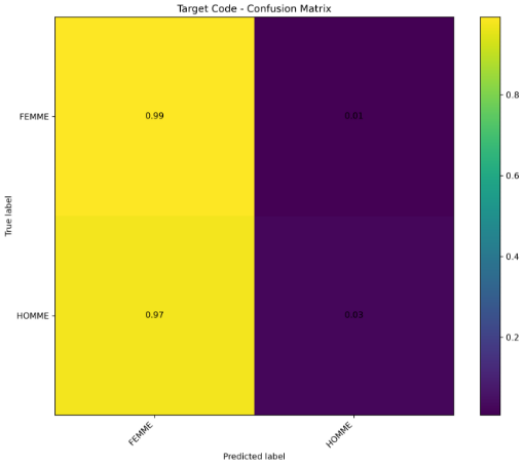
nature_code

Only a few families are recognised.



target_code

Nearly everything predicted FEMME.



The diagonal is thin: the rules recognise a handful of categories and guess the rest.

08

Why TF-IDF

Moving from counting words to weighing them

Term Frequency — TF(t, d)

How often a word appears inside one product text.

Inverse Document Frequency — IDF(t, D)

How rare that word is across the whole catalogue.

The product of both — TF-IDF = TF x IDF

A word scores high only when it is frequent here and rare elsewhere.

Why it fits this problem

Generic words neutralise themselves
plus, sans and cette appear everywhere, so IDF pushes their weight towards zero.

Rare words gain the weight
mascara, sillage or boise become decisive instead of being drowned out.

No dictionary to maintain
The weights are learned from the catalogue, not hand-written by me.

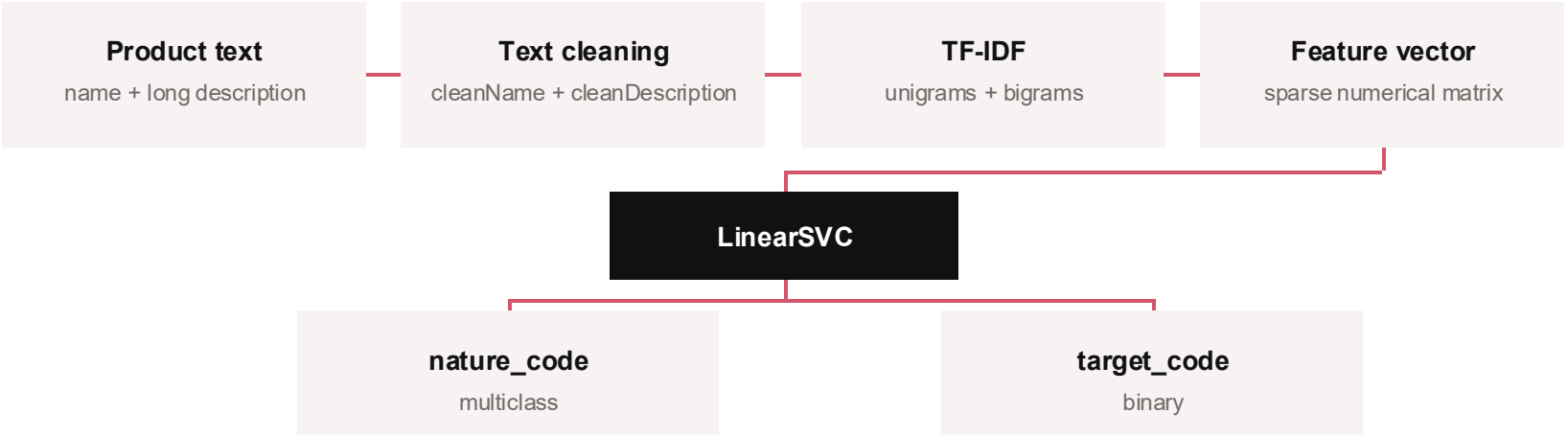
The gap the rules could not close: they count words, they cannot weigh them.

09

TF-IDF + LinearSVC Classification Pipeline

From raw product text to two predicted codes

The cleaned product name and description are converted into numerical features with TF-IDF, which captures the importance of words and word combinations across the dataset. A LinearSVC classifier then predicts nature_code and target_code.



One vectoriser, two independent classification heads.

Validation Strategy

Making the reported performance trustworthy

Stratified split

Class proportions are preserved in every fold so rare codes remain represented and results stay comparable.

Cross-validation

K-fold cross-validation on the training set, with hyper-parameters tuned only inside the folds to avoid leakage.

Honest metrics

Macro F1 alongside accuracy, plus a per-class report and confusion matrix to expose the weakest categories.

The test set is touched only once, for the final prediction file.

Results

Keyword baseline against TF-IDF + LinearSVC on the same validation set

nature_code	Rules	TF-IDF
Accuracy	0.46	**0.92**
Balanced accuracy	0.46	**0.92**
Macro F1	0.46	**0.92**

Every one of the ten categories now scores between 0.78 and 0.99 in F1.

target_code	Rules	TF-IDF
Accuracy	0.91	**0.99**
Balanced accuracy	0.51	**0.94**
Macro F1	0.50	**0.95**

HOMME rises from an F1 of 0.05 to 0.91: the minority class is finally learned.

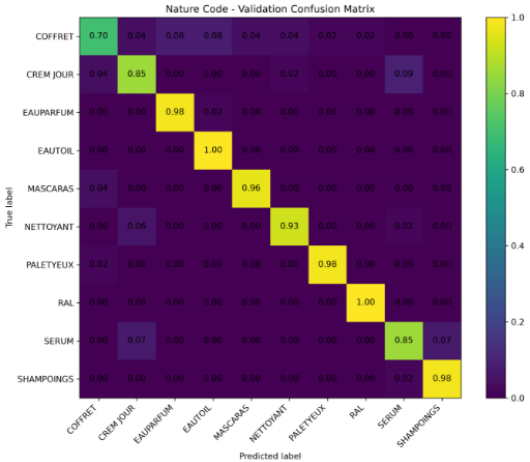
Validated on 540 held-out products with the class proportions preserved.

The Errors After Learning

Confusion matrices of TF-IDF + LinearSVC

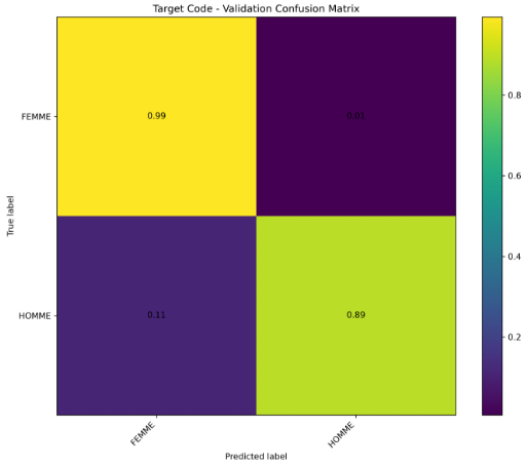
nature_code

A clean diagonal across all ten codes.



target_code

HOMME is now recognised, not ignored.



The remaining confusions are the genuinely ambiguous ones, such as gift sets and serums.

Conclusion & Improvements

What the solution delivers today and where it goes next

What works

- A simple, fully reproducible text pipeline
- Clear gain over the transparent keyword baseline
- Fast to train and cheap to run in production
- Predictions delivered in the requested format

Next steps

- Fine-tune a French language model on the descriptions
- Rebalance or reweight the rarest product codes
- Add confidence thresholds for human review
- Monitor drift as the catalogue evolves

Thank you · happy to walk through the notebook